

Lab 1: Requirements Engineering & UML Use-Case Modelling

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Problem Statement #09: Faculty Research Grant & Publication Tracker

1. Requirements Table

The table below lists five Functional Requirements (FR-001 to FR-005) and two Non-Functional Requirements (NFR-001, NFR-002) for the Faculty Research Grant & Publication Tracker system.

Faculty Research Grant & Publication Tracker — Requirements Table (Problem Statement #09)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall track grant fund allocations, allow faculty to log equipment procurement expenses, and compute the remaining budgetary balance.	High	Pass: Balance updates accurately after each expense entry and blocks expenses exceeding the grant cap. Fail: Overdraft is allowed without admin override.	Core financial control; prevents misuse of research funds.
FR-002	Functional	The system shall allow faculty to submit publication records (title, journal, DOI, indexing tier) and automatically fetch citation counts from indexed databases.	High	Pass: A submitted DOI returns verified citation count within 30 s and is stored against the faculty profile. Fail: Citation count is absent or incorrect.	Enables the dean to assess research output and impact metrics.
FR-003	Functional	The system shall enforce a co-author approval workflow where each listed co-author must digitally confirm authorship before a publication is marked 'verified'.	High	Pass: Publication status remains 'pending' until all co-authors approve; system sends reminders every 48 h. Fail: A publication is marked verified with missing approvals.	Prevents ghost authorship disputes and satisfies journal ethics requirements.
FR-004	Functional	The system shall generate a fund burn-up analytics dashboard showing cumulative expenditure vs. planned spend over the grant duration.	Medium	Pass: Dashboard reflects all logged expenses within 5 min of entry and plots a correct burn-up curve. Fail: Chart data lags by more than 5 min or plots incorrectly.	Gives the Research Dean real-time visibility into budget consumption trends.
FR-005	Functional	The system shall allow the Research Dean to configure grant milestones and trigger automated alerts when expenditure reaches 75% and 90% of the total grant amount.	Medium	Pass: Alerts are dispatched via email and in-portal notification within 1 min of crossing each threshold. Fail: Alerts are missing or delayed beyond 1 min.	Proactive oversight prevents budget overruns before the grant period ends.
NFR-001	Nonfunctional (Security & Performance)	The system shall maintain an immutable, tamper-evident audit ledger for all financial approvals and journal status modifications.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load; no record can be altered without a new versioned entry. Fail: Any record is overwritten without audit trail.	Regulatory compliance and institutional accountability for public research funds.
NFR-002	Nonfunctional (Availability)	The system shall maintain 99.5% uptime during the academic year and recover from any single-node failure within 2 minutes without data loss.	High	Pass: Monitoring logs show ≥99.5% uptime over a 30-day window; failover test completes in <2 min with zero data loss. Fail: Downtime exceeds 0.5% or recovery takes >2 min.	Faculty grant submissions and approvals are time-sensitive; downtime risks missed deadlines.

2. Requirements Critic Review

The following critique evaluates each requirement for Clarity, Testability, and Relevance, and notes any suggested improvements.

Req ID	Clarity	Testability	Relevance	Suggested Improvement
FR-001	Clear	Testable	Relevant	None
FR-002	Clear	Testable — 30 s DOI fetch is measurable	Relevant	Specify which external DBs (Scopus, Web of Science)
FR-003	Clear	Testable — status + 48 h reminder verifiable	Relevant	Define what happens if a co-author account is inactive
FR-004	Clear	Testable — 5 min lag threshold is concrete	Relevant	Specify dashboard access roles (Dean only vs Faculty too)
FR-005	Clear	Testable — 1 min alert SLA is measurable	Relevant	Add a requirement for alert channel fallback (SMS if email fails)
NFR-001	Clear	Testable — benchmark + tamper check are verifiable	Relevant	Define the retention period for the audit ledger
NFR-002	Clear	Testable — 99.5% uptime over 30-day window	Relevant	Specify RPO (Recovery Point Objective) alongside RTO

2.1 Overall Critique Summary

All seven requirements are clearly written using the 'The system shall...' convention, making them unambiguous and role-specific. Every acceptance criterion offers a concrete pass/fail condition, making automated test design straightforward. The requirements are tightly scoped to the problem domain (grant tracking, publication metrics, co-author workflow). Minor improvements are recommended for FR-002 (naming specific citation databases), FR-005 (alert channel fallback), NFR-001 (ledger retention period), and NFR-002 (adding an RPO alongside the stated RTO).

3. UML Use-Case Diagram

Actors: Faculty Researcher, Research Dean, External Citation DB

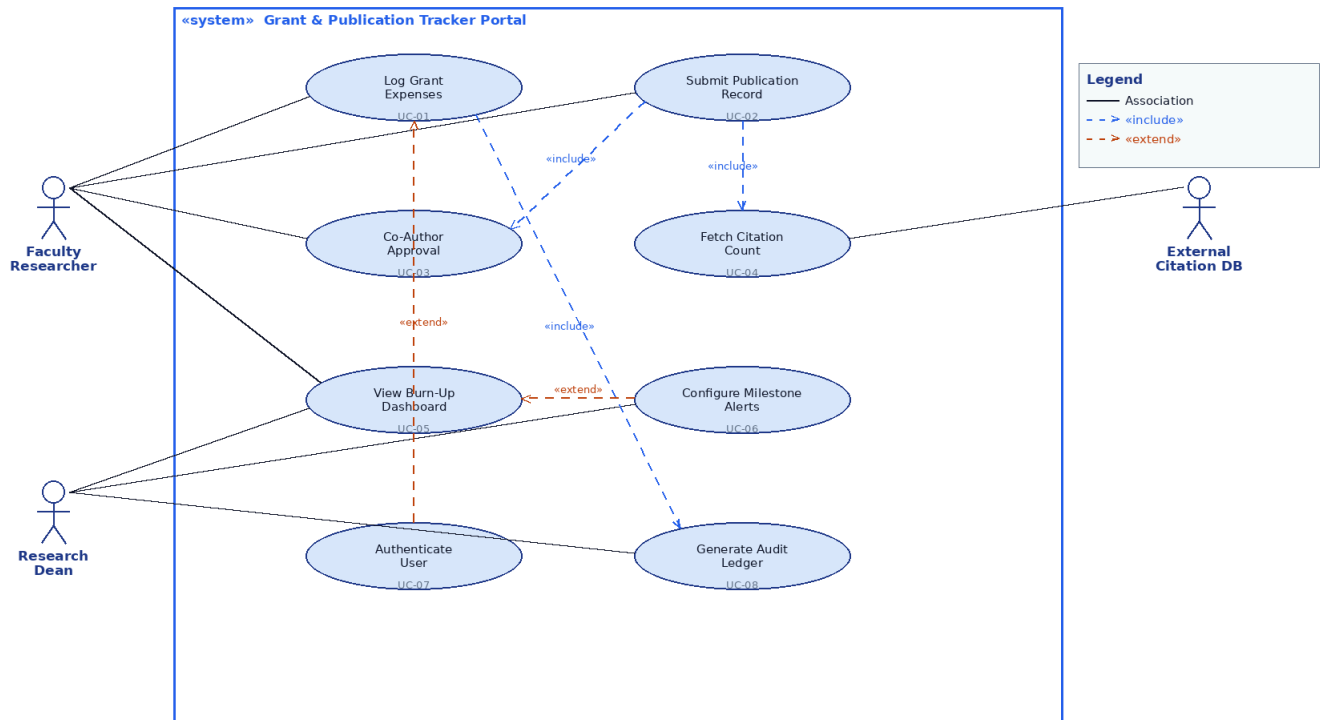
Use Cases: UC-01 Log Grant Expenses, UC-02 Submit Publication Record, UC-03 Co-Author Approval, UC-04 Fetch Citation Count, UC-05 View Burn-Up Dashboard, UC-06 Configure Milestone Alerts, UC-07 Authenticate User, UC-08 Generate Audit Ledger

«include» relationships: UC-02 → UC-04 (Submit Publication always fetches citation count); UC-02 → UC-03 (Submit Publication always triggers co-author approval); UC-01 → UC-08 (Log Expenses always writes to audit ledger).

«extend» relationships: UC-06 → UC-05 (Configure Alerts optionally extends View Dashboard); UC-07 → UC-01 (Authenticate optionally called for sensitive grant operations).

UML Use-Case Diagram — Faculty Research Grant & Publication Tracker

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4. Use-Case Flow Specification

The flow specification below covers UC-02: Submit Publication Record, which is the core use case of the system.

Use Case ID: UC-02

Use Case: Submit Publication Record

Actors: Faculty Researcher (primary), External Citation DB (secondary)

Preconditions: 1. Faculty Researcher is authenticated. 2. At least one active grant record exists. 3. The publication DOI is valid and resolvable.

Postconditions: 1. Publication record stored with status 'verified'. 2. Citation count recorded against faculty profile. 3. Immutable audit ledger entry created.

4.1 Main Success Scenario

1. Faculty Researcher selects "Submit Publication" from the portal dashboard.
2. System displays the publication submission form (title, journal, DOI, indexing tier, co-author list).
3. Faculty fills in all required fields and clicks "Submit".
4. System validates the form: checks mandatory fields and DOI format.
5. System sends the DOI to the External Citation DB and retrieves the citation count within 30 s.
6. System stores the publication record with status "pending" and attaches the citation count.
7. System dispatches co-author confirmation emails with a one-click approval link.
8. Each co-author clicks the approval link and confirms authorship.
9. Once all co-authors have approved, the system updates publication status to "verified".
10. System creates an immutable audit ledger entry recording the submission and all approvals.
11. System displays a confirmation notification to the Faculty Researcher. Use case ends successfully.

4.2 Alternate Flows

4a — Invalid DOI Format

- 4a1. System highlights the DOI field: "Invalid DOI format."
- 4a2. Faculty corrects the DOI and resubmits.
- 4a3. Flow returns to Step 4.

5a — External Citation DB Unreachable

- 5a1. System retries the citation fetch up to 3 times with a 10 s interval.
- 5a2. If all retries fail, system stores the record with citation count = N/A and flags it for manual update.
- 5a3. System notifies the Research Dean.
- 5a4. Flow continues from Step 6.

8a — Co-Author Does Not Respond Within 48 h

- 8a1. System sends a reminder email to the non-responding co-author.
- 8a2. If no response after 96 h, system escalates to the Research Dean for manual override.
- 8a3. Research Dean approves or rejects on behalf of the co-author.
- 8a4. Flow returns to Step 9.